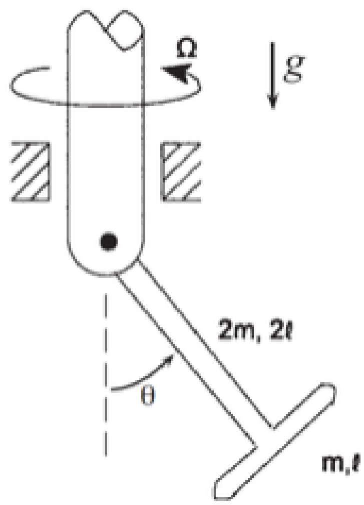
Problem 1

Problem 1



A bar of mass m, length ℓ is
welded to a bar of mass $2m$, length 2ℓ to form the T shown. The body is pinned to a post that rotates at the constant rate Ω.

The moment of inertia tensor for the T-shaped bar is given by,

$$I = \begin{bmatrix} \frac{1}{12}m\ell^2 & 0 & 0 \\ 0 & \frac{20}{3}m\ell^2 & 0 \\ 0 & 0 & \frac{27}{4}m\ell^2 \end{bmatrix}$$

in the body frame $\{\hat{b}\}$ along the principal axes of the body, centered at the pivot point, with $\hat{b}_1$ pointing along the length-2ℓ bar (down and to the right), $\hat{b}_2$ pointing up and to the right parallel to the length-ℓ bar, and $\hat{b}_3$ pointing out of the page.

(a) Use the vector differentiation formula to write the velocity of each particle.

$$\hat{a}_1 = \cos\phi\,\hat{n}_1 + \sin\phi\,\hat{n}_2, \qquad \hat{a}_3 = \hat{n}_3, \qquad \hat{a}_2 = \hat{a}_3 \times \hat{a}_1, \qquad {}^N\vec{\omega}^A = \dot{\phi}\,\hat{n}_3$$

$$\hat{e}_{r_1} = \cos\theta\,\hat{a}_1 - \sin\theta\,\hat{a}_3, \qquad \hat{e}_{\theta_1} = -\sin\theta\,\hat{a}_1 - \cos\theta\,\hat{a}_3 \qquad (\text{mirrored: } \hat{e}_{r_2}, \hat{e}_{\theta_2})$$

$${}^N\vec{\omega}^{E_1} = \dot{\phi}\,\hat{a}_3 + \dot{\theta}\,\hat{a}_2, \qquad {}^N\vec{\omega}^{E_2} = \dot{\phi}\,\hat{a}_3 - \dot{\theta}\,\hat{a}_2$$

$$\vec{r}_1 = \ell\hat{e}_{r_1}, \qquad \vec{r}_2 = \ell\hat{e}_{r_2}, \qquad \vec{r}_M = -2\ell\sin\theta\,\hat{n}_3$$

$$\begin{aligned} {}^N\dot{\vec{r}}_1 &= {}^N\vec{\omega}^{E_1} \times \vec{r}_1 = \left(\dot{\phi}\,\hat{a}_3 + \dot{\theta}\,\hat{a}_2\right) \times (\ell\cos\theta\,\hat{a}_1 - \ell\sin\theta\,\hat{a}_3) \\ &= \ell\cos\theta\,\dot{\phi}\,\hat{a}_2 - \ell\cos\theta\,\dot{\theta}\,\hat{a}_3 - \ell\sin\theta\,\dot{\theta}\,\hat{a}_1 = \ell\cos\theta\,\dot{\phi}\,\hat{a}_2 + \ell\dot{\theta}\,\hat{e}_{\theta_1} \end{aligned}$$

$$\boxed{\vec{V}_1 = \ell\cos\theta\,\dot{\phi}\,\hat{a}_2 + \ell\dot{\theta}\,\hat{e}_{\theta_1}} \qquad (\text{symmetry}) \qquad \boxed{\vec{V}_2 = -\ell\cos\theta\,\dot{\phi}\,\hat{a}_2 + \ell\dot{\theta}\,\hat{e}_{\theta_2}}$$

$$\boxed{\vec{V}_M = \frac{d\vec{r}_M}{dt} = -2\ell\cos\theta\,\dot{\theta}\,\hat{n}_3}$$

(b) What is the minimal set of generalized coordinates of the system? What is a redundant set that you could consider?

$$\text{minimal: } \boxed{\{\theta,\ \phi\}} \qquad \text{redundant: } \boxed{\{\theta_1,\ \phi_1,\ \theta_2,\ \phi_2,\ z_M\}}$$

(c) Using the minimal set of coordinates, calculate the generalized velocity, $\dot{\vec{r}}_{ij}$, for that generalized coordinate as if you were solving this problem with D'Alembert's Principle (but don't work through the entire solution).

$$\vec{V}_1 = \ell\dot{\theta}\,\hat{e}_{\theta_1} + \ell\cos\theta\,\dot{\phi}\,\hat{a}_2, \qquad \vec{V}_2 = \ell\dot{\theta}\,\hat{e}_{\theta_2} - \ell\cos\theta\,\dot{\phi}\,\hat{a}_2, \qquad \vec{V}_M = -2\ell\cos\theta\,\dot{\theta}\,\hat{n}_3$$

$$\boxed{\gamma_{1\theta} = \ell\,\hat{e}_{\theta_1}}, \quad \boxed{\gamma_{2\theta} = \ell\,\hat{e}_{\theta_2}}, \quad \boxed{\gamma_{M\theta} = -2\ell\cos\theta\,\hat{n}_3}$$

$$\boxed{\gamma_{1\phi} = \ell\cos\theta\,\hat{a}_2}, \quad \boxed{\gamma_{2\phi} = -\ell\cos\theta\,\hat{a}_2}, \quad \boxed{\gamma_{M\phi} = \vec{0}}$$

(d) Construct the Lagrangian and derive the equations of motion for each generalized coordinate q_i of the system.

$$|\vec{V}_1|^2 = |\vec{V}_2|^2 = \ell^2\dot{\theta}^2 + \ell^2\cos^2\theta\,\dot{\phi}^2, \qquad |\vec{V}_M|^2 = 4\ell^2\cos^2\theta\,\dot{\theta}^2$$

$$T = m\ell^2\dot{\theta}^2 + m\ell^2\cos^2\theta\,\dot{\phi}^2 + 2M\ell^2\cos^2\theta\,\dot{\theta}^2, \qquad V = -2(m+M)g\ell\sin\theta$$

$$\boxed{\mathcal{L} = T - V = \ell^2\left(m + 2M\cos^2\theta\right)\dot{\theta}^2 + m\ell^2\cos^2\theta\,\dot{\phi}^2 + 2(m+M)g\ell\sin\theta}$$

$$\begin{aligned}
\frac{\partial\mathcal{L}}{\partial\dot{\theta}} &= 2\ell^2\left(m + 2M\cos^2\theta\right)\dot{\theta} \\
\frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{\theta}}\right) &= 2\ell^2\left(m + 2M\cos^2\theta\right)\ddot{\theta} - 8M\ell^2\dot{\theta}^2\sin\theta\cos\theta \\
\frac{\partial\mathcal{L}}{\partial\theta} &= -4M\ell^2\dot{\theta}^2\sin\theta\cos\theta - 2m\ell^2\dot{\phi}^2\sin\theta\cos\theta + 2(m+M)g\ell\cos\theta
\end{aligned}$$

$$\boxed{(m + 2M\cos^2\theta)\ddot{\theta} - 2M\dot{\theta}^2\sin\theta\cos\theta + m\dot{\phi}^2\sin\theta\cos\theta - (m+M)\frac{g}{\ell}\cos\theta = 0}$$

ϕ cyclic ($\partial\mathcal{L}/\partial\phi = 0$):

$$2m\ell^2\cos^2\theta\,\ddot{\phi} - 4m\ell^2\dot{\theta}\dot{\phi}\sin\theta\cos\theta = 0 \quad \Rightarrow \quad \boxed{\ddot{\phi} = 2\dot{\theta}\dot{\phi}\tan\theta}$$

(e) Construct the Hamiltonian of the system and write the Hamiltonian equations of motion for each generalized coordinate q_i and its conjugate momentum p_i.

$$p_\theta = 2\ell^2(m + 2M\cos^2\theta)\dot{\theta}, \qquad p_\phi = 2m\ell^2\cos^2\theta\,\dot{\phi}$$

$$\boxed{H = T + V = \frac{p_\theta^2}{4\ell^2(m + 2M\cos^2\theta)} + \frac{p_\phi^2}{4m\ell^2\cos^2\theta} - 2(m+M)g\ell\sin\theta}$$

Hamilton's equations of motion:

$$\boxed{\dot{\theta} = \frac{p_\theta}{2\ell^2(m + 2M\cos^2\theta)}} \qquad \boxed{\dot{\phi} = \frac{p_\phi}{2m\ell^2\cos^2\theta}}$$

$$\boxed{\dot{p}_\theta = -\frac{Mp_\theta^2 \sin\theta\cos\theta}{\ell^2(m + 2M\cos^2\theta)^2} - \frac{p_\phi^2 \sin\theta}{2m\ell^2\cos^3\theta} + 2(m + M)g\ell\cos\theta}$$

$$\boxed{\dot{p}_\phi = 0}$$

(f) Are there any constants of motion for this problem? List them.

Two constants of motion:

1. $\boxed{p_\phi = 2m\ell^2\cos^2\theta\,\dot{\phi}}$, the angular momentum of the two balls about the shaft axis: ϕ is cyclic, so no external torque acts about $\hat{n}_3$.
2. $\boxed{H = E}$, the total mechanical energy: $\partial H/\partial t = 0$, and therefore $H = T + V$.

(g) Choose your own parameters and initial conditions for the problem. Using an initial value solver, numerically integrate Hamilton's equations of motion, and plot each q_i, p_i vs time. Additionally, plot θ vs. $\dot{\phi}$.

Parameters: $m = 1$, $M = 2$, $\ell = 1$; initial conditions $(\theta_0, \phi_0, p_{\theta_0}, p_{\phi_0}) = (0.3,\ 0,\ 0,\ 3)$ (SI-consistent units).

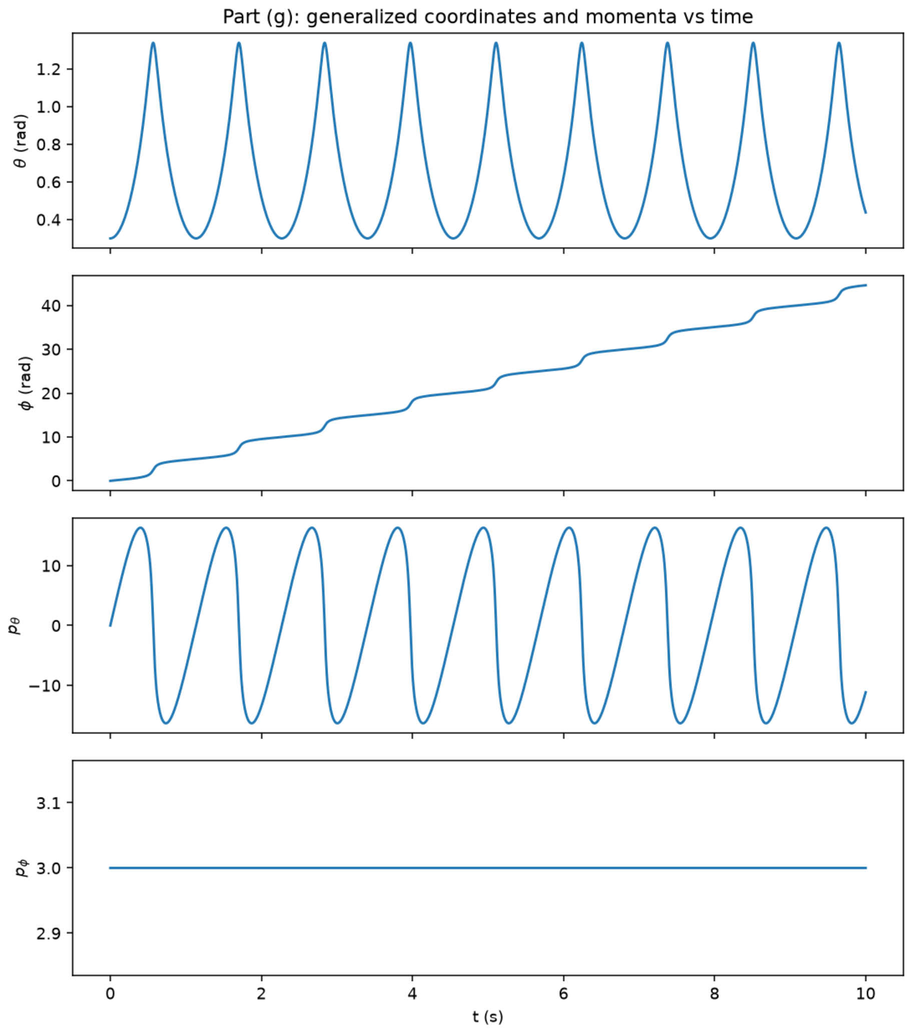


Figure 1: Part (g): generalized coordinates and momenta vs. time.

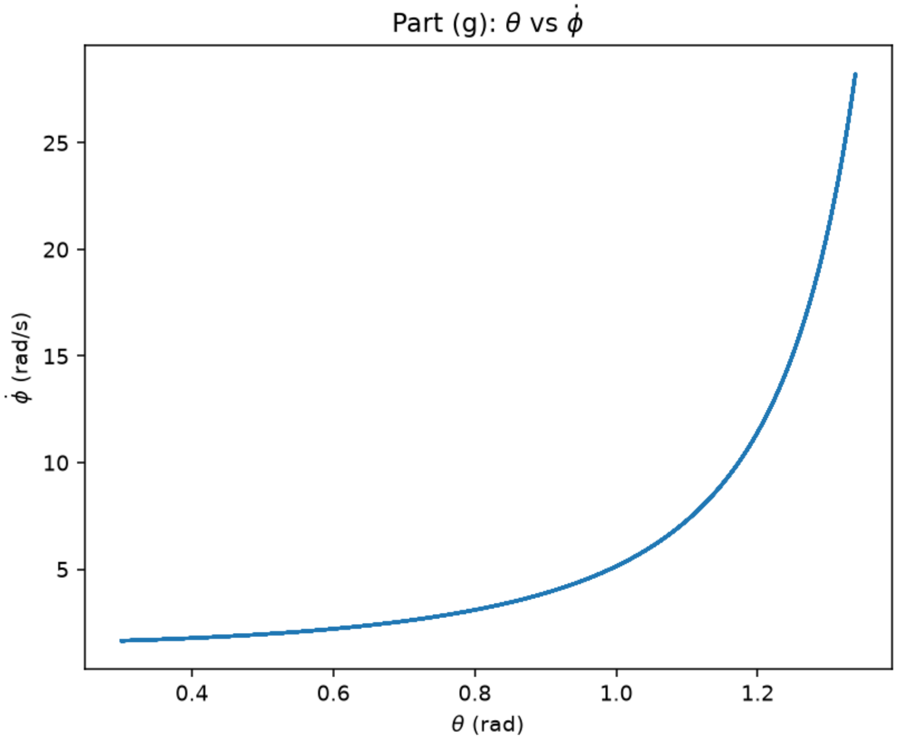


Figure 2: Part (g): θ vs. $\dot{\phi}$.

(h) If you wanted to solve for the forces in each bar to make sure your string was strong enough, how could you solve for that?

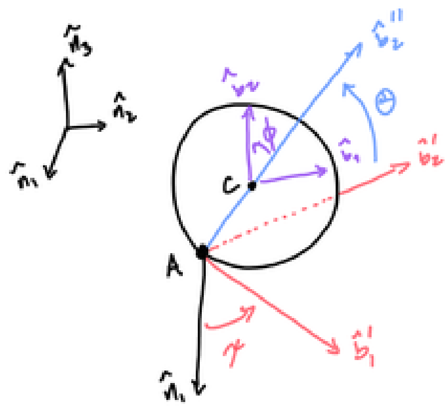
Newton's second law on each of the three masses gives three vector equations; the constraint force is whatever is left over once we plug in the accelerations from the Lagrangian or Hamiltonian solution. A Lagrange-multiplier approach gets there more directly, folding the bar force into the equations of motion from the start at the cost of more work.

(i) Discuss some possible approaches to the problem, including the Newtonian approach, which I didn't ask you to use here. Which approaches do you prefer? What are the different insights gained from each one?

Any of the approaches from class would work here. Newtonian mechanics, $\vec{F} = m\vec{a}$, gets out of hand really fast with three whole particles and their constraint forces. I prefer the Lagrangian approach because everything can be put straight into the Euler–Lagrange equation. The Hamiltonian formulation splits the system into first-order equations, which is better for the numerical solver. All three, including D'Alembert's principle, discard the constraint forces along the way, so recovering them later takes extra work.

Problem 2

Problem 2



Consider a coin rolling along its edge on a flat surface. To describe its orientation, we consider the position of the center of the coin to be defined as,

$$\vec{r}_c = x\hat{n}_1 + y\hat{n}_2 + z\hat{n}_3.$$

To describe its orientation, we use 3-1-3 Euler angles, just as in HW6. Consider the Euler angles to be:

First, imagine the coin to be laying flat on the ground. Rotate the heading of the coin by a counterclockwise angle ψ around the $\hat{n}_3$ axis. Second, rotate the coin upward around the $\hat{b}_1'$ axis by the angle θ. If $\theta = 90^\circ$, the coin is standing on its edge, pointing in the direction of $\hat{b}_1'$. Finally, we consider rolling around the $\hat{b}_3 = \hat{b}_3''$ axis by the angle ϕ.

NOTE: If you're having a hard time understanding the orientation of the coin, feel free to reach out! I'll try to re-describe its orientation.

(a) Write the two equations of constraint for the knife-edge constraints at A ($\phi_1(q,\dot{q}) = 0$) and C ($\phi_2(q,\dot{q}) = 0$). Hint: you may be able to use ϕ_1 to eliminate some terms from ϕ_2.

$$\hat{b}_1 = \cos\theta\,\hat{n}_1 + \sin\theta\,\hat{n}_2, \quad \hat{b}_2 = -\sin\theta\,\hat{n}_1 + \cos\theta\,\hat{n}_2$$
$$\hat{c}_1 = \cos(\theta+\psi)\,\hat{n}_1 + \sin(\theta+\psi)\,\hat{n}_2, \quad \hat{c}_2 = -\sin(\theta+\psi)\,\hat{n}_1 + \cos(\theta+\psi)\,\hat{n}_2$$

At A: ${}^N\vec{V}_A = \dot{x}\,\hat{n}_1 + \dot{y}\,\hat{n}_2$, knife edge $\Rightarrow {}^N\vec{V}_A \cdot \hat{b}_2 = 0$:

$$\boxed{\phi_1 = -\dot{x}\sin\theta + \dot{y}\cos\theta = 0}$$

At C: $\vec{r}_C = \vec{r}_A + d\,\hat{b}_1 - \ell\,\hat{c}_1$, ${}^N\vec{\omega}^B = \dot{\theta}\,\hat{n}_3$, ${}^N\vec{\omega}^C = (\dot{\theta}+\dot{\psi})\hat{n}_3$

$$\frac{{}^N d\hat{b}_1}{dt} = \dot{\theta}\,\hat{b}_2, \qquad \frac{{}^N d\hat{c}_1}{dt} = (\dot{\theta}+\dot{\psi})\,\hat{c}_2$$

$${}^N\vec{V}_C = {}^N\vec{V}_A + d\dot{\theta}\,\hat{b}_2 - \ell(\dot{\theta}+\dot{\psi})\,\hat{c}_2$$

Knife edge $\Rightarrow \vec{V}_C \cdot \hat{c}_2 = 0$; using $\hat{b}_2 \cdot \hat{c}_2 = \cos\psi$:

$$^N\vec{V}_A \cdot \hat{c}_2 + d\dot{\theta}\cos\psi - \ell(\dot{\theta} + \dot{\psi}) = 0$$

$$^N\vec{V}_A \cdot \hat{c}_2 = \cos\psi\,\underbrace{(-\dot{x}\sin\theta + \dot{y}\cos\theta)}_{=\,\phi_1\,=\,0} - \sin\psi(\dot{x}\cos\theta + \dot{y}\sin\theta) = -\sin\psi(\dot{x}\cos\theta + \dot{y}\sin\theta)$$

$$\boxed{\phi_2 = -\sin\psi(\dot{x}\cos\theta + \dot{y}\sin\theta) + (d\cos\psi - \ell)\dot{\theta} - \ell\dot{\psi} = 0}$$

(b) Use Lagrange's equations to derive the equations of motion for the cart in terms of Lagrange multipliers λ_1 and λ_2 corresponding to the two constraints.

$$V = 0, \qquad \mathcal{L} = T = \tfrac{1}{2}M(\dot{x}^2 + \dot{y}^2) + \tfrac{1}{2}I\dot{\theta}^2$$

Pfaffian form $\phi_j = \sum_i a_{ji}\dot{q}_i + b_j = 0$, $q = (x, y, \theta)$ ($\psi(t)$ given):

$$a_{1x} = -\sin\theta, \quad a_{1y} = \cos\theta, \quad a_{1\theta} = 0$$

$$a_{2x} = -\sin\psi\cos\theta, \quad a_{2y} = -\sin\psi\sin\theta, \quad a_{2\theta} = d\cos\psi - \ell$$

$$\boxed{\begin{aligned} M\ddot{x} &= -\lambda_1\sin\theta - \lambda_2\sin\psi\cos\theta \\ M\ddot{y} &= \lambda_1\cos\theta - \lambda_2\sin\psi\sin\theta \\ I\ddot{\theta} &= \lambda_2\,(d\cos\psi - \ell) \end{aligned}}$$

(c) Solve for λ_1 and λ_2.

Let $v \equiv {}^N\vec{V}_A \cdot \hat{b}_1$. With $\phi_1 = 0$: $\dot{x} = v\cos\theta,\ \dot{y} = v\sin\theta \ \Rightarrow\ {}^N\vec{a}_A = \dot{v}\,\hat{b}_1 + v\dot{\theta}\,\hat{b}_2$.

$$M\,{}^N\vec{a}_A = \lambda_1\hat{b}_2 - \lambda_2\sin\psi\,\hat{b}_1 \quad\Rightarrow\quad M\dot{v} = -\lambda_2\sin\psi, \qquad \lambda_1 = Mv\dot{\theta}$$

$\phi_2 = 0$ in terms of v: $-v\sin\psi + (d\cos\psi - \ell)\dot{\theta} - \ell\dot{\psi} = 0 \ \Rightarrow$

$$\dot{\theta} = \frac{\ell\dot{\psi} + v\sin\psi}{d\cos\psi - \ell} \qquad\Rightarrow\qquad \boxed{\lambda_1 = Mv\,\frac{\ell\dot{\psi} + v\sin\psi}{d\cos\psi - \ell}}$$

Differentiating $\dot{\theta}$ and eliminating $\dot{v}$ via $M\dot{v} = -\lambda_2\sin\psi$, $\lambda_2 = I\ddot{\theta}/(d\cos\psi - \ell)$:

$$\ddot{\theta} = \frac{(d\cos\psi - \ell)(v\dot{\psi}\cos\psi + \ell\ddot{\psi}) + d\dot{\psi}\sin\psi(v\sin\psi + \ell\dot{\psi})}{(d\cos\psi - \ell)^2 + \dfrac{I}{M}\sin^2\psi} \qquad\Rightarrow\qquad \boxed{\lambda_2 = \frac{I}{d\cos\psi - \ell}\,\ddot{\theta}}$$

(d) Assume the rider rotates the front wheel back and forth with $\psi = \frac{\pi}{6}\cos(\omega_0 t)$ rad. Plot a trajectory of x vs y (the motion of the center of mass) as well as θ vs t, given that the Crazy Cart starts at rest with $\theta_0 = 0$. You may choose your own values for mass, lengths, and moment of inertia.

Parameters: $M = 1$, $I = 0.5$, $d = 1$, $\ell = 0.5$, $\omega_0 = 1$ (SI-consistent units).

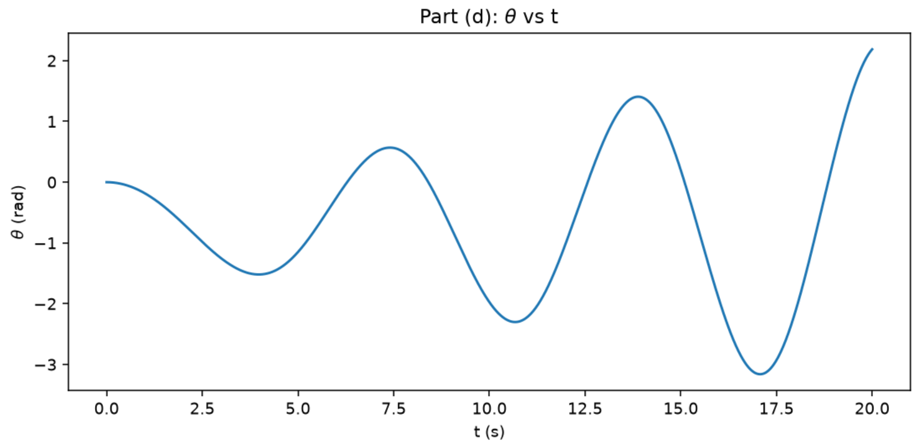


Figure 3: Part (d): θ vs. t.

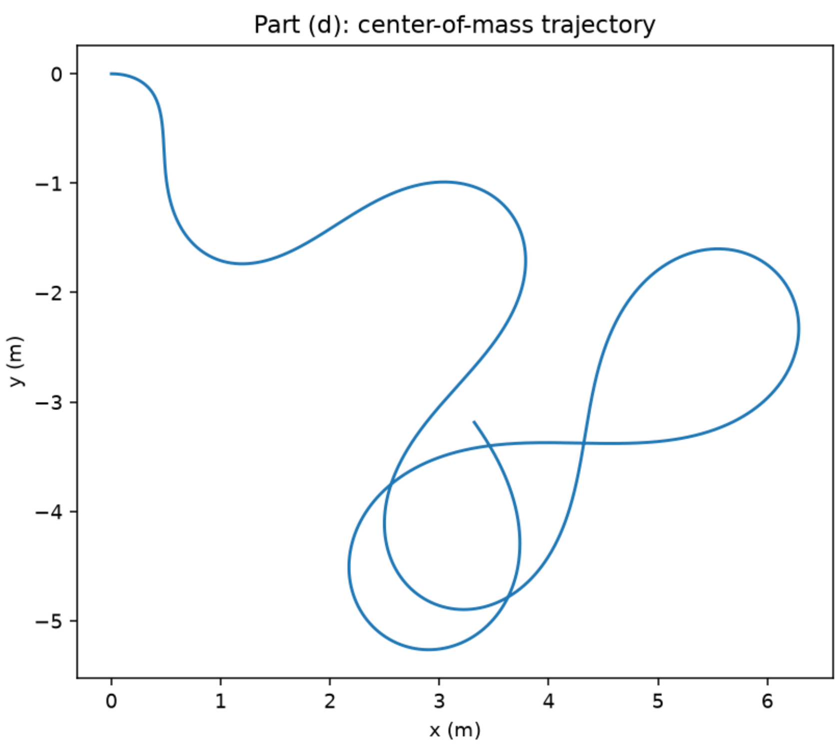


Figure 4: Part (d): center-of-mass trajectory under prescribed steering $\psi(t)$.

(e) What are the key assumptions that we have made in deriving this system? What practicalities of real systems would violate these assumptions? What effect do you expect that would have on the dynamics?

Neglecting the front wheel's mass is the biggest one. It simplifies the kinetic energy and makes the whole system driven by the prescribed steering angle $\psi(t)$: the equations never solve for ψ, the rider just feeds it in as a function of time. If we keep that mass term instead, and ψ needs its own equation of motion, with its own inertia and angular momentum, which would couple through the constraint force λ_2 at C into the cart's translational and rotational equations. The rider's steering input would then show up as a torque acting back on the cart.

We also ignored friction and slipping at the wheels. Friction adds a non-conservative force to the energy balance. Slipping breaks the knife-edge constraints $\phi_1 = 0$ and $\phi_2 = 0$ straight up, giving the cart an extra degree of freedom in the lateral direction for as long as it lasts. The model holds only as long as the ground supplies enough friction to keep both wheels tracking. Once the wheels slip, the real motion diverges from the nonholonomic prediction.

Appendix: Code Listings

Problem 1 Code

```python
import os
import numpy as np
from scipy.integrate import solve_ivp
import matplotlib.pyplot as plt

FIG_DIR = os.path.join(os.path.dirname(__file__), '..', 'latex', 'Figures', 'Graphs', 'Problem1')
os.makedirs(FIG_DIR, exist_ok=True)

g = 9.81


def hamilton_deriv(t, y, m, M, L, g):
    theta, phi, p_theta, p_phi = y
    c, s = np.cos(theta), np.sin(theta)
    theta_dot = p_theta / (2*L**2*(m + 2*M*c**2))
    phi_dot = p_phi / (2*m*L**2*c**2)
    p_theta_dot = (-M*p_theta**2*s*c/(L**2*(m + 2*M*c**2)**2)
                   - p_phi**2*s/(2*m*L**2*c**3)
                   + 2*(m+M)*g*L*c)
    return [theta_dot, phi_dot, p_theta_dot, 0.0]

m, M, L = 1.0, 2.0, 1.0
theta0, phi0, ptheta0, pphi0 = 0.3, 0.0, 0.0, 3.0
t_end_g = 10.0
t_eval_g = np.linspace(0, t_end_g, 3000)

sol = solve_ivp(hamilton_deriv, [0, t_end_g], [theta0, phi0, ptheta0, pphi0],
                args=(m, M, L, g), t_eval=t_eval_g, method='DOP853', rtol=1e-10, atol=1e-12)
theta, phi, p_theta, p_phi = sol.y
phi_dot = p_phi / (2*m*L**2*np.cos(theta)**2)

H = (p_theta**2/(4*L**2*(m + 2*M*np.cos(theta)**2)) + p_phi**2/(4*m*L**2*np.cos(theta)**2)
     - 2*(m+M)*g*L*np.sin(theta))
print(f"max relative drift in H: {np.max(np.abs(H - H[0]))/abs(H[0]):.3e}")

fig, ax = plt.subplots(4, 1, sharex=True, figsize=(8, 9))
ax[0].plot(sol.t, theta); ax[0].set_ylabel(r'$\theta$ (rad)')
ax[1].plot(sol.t, phi); ax[1].set_ylabel(r'$\phi$ (rad)')
ax[2].plot(sol.t, p_theta); ax[2].set_ylabel(r'$p_\theta$')
ax[3].plot(sol.t, p_phi); ax[3].set_ylabel(r'$p_\phi$'); ax[3].set_xlabel('t (s)')
ax[0].set_title('Part (g): generalized coordinates and momenta vs time')
fig.tight_layout(); fig.savefig(os.path.join(FIG_DIR, 'partG_qp_vs_t.png'), dpi=140)

fig, ax = plt.subplots(figsize=(6, 5))
ax.plot(theta, phi_dot)
ax.set_xlabel(r'$\theta$ (rad)'); ax.set_ylabel(r'$\dot\phi$ (rad/s)')
ax.set_title(r'Part (g): $\theta$ vs $\dot\phi$')
fig.tight_layout(); fig.savefig(os.path.join(FIG_DIR, 'partG_theta_vs_phidot.png'), dpi=140)

plt.show()
```

Listing 1: Numerical integration of Hamilton's equations of motion for the flyball governor, with an energy-conservation check.

Problem 2 Code

```python
import os
import numpy as np
from scipy.integrate import solve_ivp
import matplotlib.pyplot as plt

FIG_DIR = os.path.join(os.path.dirname(__file__), '..', 'latex', 'Figures', 'Graphs', 'Problem2')
os.makedirs(FIG_DIR, exist_ok=True)


def deriv(t, state, M, I, d, l, omega0):
    v, theta, x, y = state
    psi = (np.pi/6)*np.cos(omega0*t)
    psi_dot = -(np.pi/6)*omega0*np.sin(omega0*t)
    psi_ddot = -(np.pi/6)*omega0**2*np.cos(omega0*t)
    c, s = np.cos(psi), np.sin(psi)
    denom = d*c - l

    theta_dot = (l*psi_dot + v*s)/denom
    theta_ddot = ((denom*(v*psi_dot*c + l*psi_ddot) + d*psi_dot*s*(v*s + l*psi_dot))
                  / (denom**2 + (I/M)*s**2))
    v_dot = -I*theta_ddot*s/(M*denom)

    return [v_dot, theta_dot, v*np.cos(theta), v*np.sin(theta)]

M, I, d, l, omega0 = 1.0, 0.5, 1.0, 0.5, 1.0
t_end = 20.0
t_eval = np.linspace(0, t_end, 4000)

sol = solve_ivp(deriv, [0, t_end], [0.0, 0.0, 0.0, 0.0], args=(M, I, d, l, omega0),
                t_eval=t_eval, method='DOP853', rtol=1e-10, atol=1e-12, dense_output=True)
v, theta, x, y = sol.y

t_f = np.linspace(0, t_end, 200001)
v_f, th_f, x_f, y_f = sol.sol(t_f)
psi_f, psid_f = (np.pi/6)*np.cos(omega0*t_f), -(np.pi/6)*omega0*np.sin(omega0*t_f)
xd_f, yd_f, thd_f = np.gradient(x_f, t_f), np.gradient(y_f, t_f), np.gradient(th_f, t_f)
phi1 = -xd_f*np.sin(th_f) + yd_f*np.cos(th_f)
phi2 = -np.sin(psi_f)*(xd_f*np.cos(th_f)+yd_f*np.sin(th_f)) + (d*np.cos(psi_f)-l)*thd_f - l*psid_f
print(f"max |phi1|, |phi2|: {np.max(np.abs(phi1)):.3e}, {np.max(np.abs(phi2)):.3e}")

fig, ax = plt.subplots(figsize=(6, 6))
ax.plot(x, y)
ax.set_xlabel('x (m)'); ax.set_ylabel('y (m)'); ax.set_aspect('equal')
ax.set_title('Part (d): center-of-mass trajectory')
fig.tight_layout(); fig.savefig(os.path.join(FIG_DIR, 'partD_xy_trajectory.png'), dpi=140)

fig, ax = plt.subplots(figsize=(8, 4))
ax.plot(sol.t, theta)
ax.set_xlabel('t (s)'); ax.set_ylabel(r'$\theta$ (rad)')
ax.set_title(r'Part (d): $\theta$ vs t')
fig.tight_layout(); fig.savefig(os.path.join(FIG_DIR, 'partD_theta_vs_t.png'), dpi=140)

plt.show()
```

Listing 2: Numerical integration of the Crazy Cart's equations of motion under prescribed steering $\psi(t)$, with a constraint-satisfaction check.